

AST 6007-01 Spacecraft Dynamics and Control

Homework #5

Due Dec 1 2021, 10:00AM

A rigid spacecraft in a circular orbit at an altitude of 300km has principal moments of inertia $I_x = 1,400kg - m^2$, $I_y = 1,700kg - m^2$, $I_z = 1,000kg - m^2$. It is desired to perform a rest-to-rest maneuver of the spacecraft from an initial attitude given by Euler angles $\psi = 0.5, \theta = 0.3, \phi = 0.0$ radians to a terminal attitude given by Euler angles $\psi = 0, \theta = 0, \phi = 0$ radians. The slewing maneuver is to be completed in exactly 120 seconds.

1. Include the gravity gradient moments and make use of the nonlinear rotational equations of motion for a spacecraft in a circular orbit. Assume the slewing maneuver is to be performed by rotating the spacecraft about a constant axis.
 - a. Determine the constant axis of rotation for the slewing maneuver. What are the initial and final values of the rotation angle about that axis?
 - b. Determine roll-axis, pitch-axis and yaw-axis control torques that achieve the slewing maneuver. Plot the time responses for the three control torques and for the three Euler angles.
2. Include the gravity gradient moments and make use of the linearized rotational equations of motion for a spacecraft in a circular orbit.
 - a. Determine a two impulse solution for the pitch-axis control torque that achieves the pitch-axis slewing maneuver. What is the change in the pitch angular velocity component at each impulse? Plot the time response for the pitch angle.
 - b. Determine a two impulse solution for the roll-axis and yaw-axis control torques that achieve the roll-yaw-axes slewing maneuver. What are the changes in roll and yaw angular velocity components at each impulse? Plot the time responses for the roll angle and the yaw angle.
 - c. Determine a piecewise constant pitch-axis control torque that achieves the pitch-axis slewing maneuver. Plot the time response for the pitch angle.
 - d. Determine piecewise constant roll-axis and yaw-axis control torques that achieve the roll-yaw-axes slewing maneuver. Plot the time responses for the roll angle and the yaw angle.
 - e. Why is it a good engineering approximation to use the linearized equations to determine the control torques in parts a, b, c and d?
3. Compare the strengths and weaknesses of your three different solutions to the rest-to-rest slewing maneuver problem:
 - The nonlinear control approach in Problem 1
 - The linear control approach in Problem 2, parts a and b
 - The linear control approach in Problem 2, parts c and d